" OrbitMind AI– Intelligent Productivity & Study Companion"

by

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CHAPTER I

Introduction

1.1 Project Title

OrbitMind AI – Intelligent Productivity & Study Companion

1.2 Project Overview

OrbitMind AI is a comprehensive, full-stack web application engineered to serve as a highly personalized academic assistant for university students. By integrating modern web technologies (Node.js, Express, HTML/JS) with advanced artificial intelligence models (Google Gemini / OpenAI), the platform bridges the gap between traditional task management and intelligent, context-aware academic coaching. It provides an intuitive dashboard to manage workloads, an AI-powered engine to generate structured study roadmaps from raw syllabi, a smart note summarizer, and a 24/7 conversational AI coach that converts natural language directly into actionable system tasks.

1.3 Problem Statement

University students frequently struggle to balance overlapping assignment deadlines, exam preparations, and personal commitments, leading to heightened academic stress and burnout. Traditional productivity applications are rigid; they require heavy manual input and lack a contextual understanding of a student's actual cognitive load. They do not comprehend exam proximities, topic complexities, or when a student is at risk of overworking. Consequently, students spend excessive time organizing their work rather than effectively studying, resulting in decreased productivity and inconsistent academic performance.

1.4 Objectives

- **Assist students with dynamic study planning:** Automatically generate day-by-day study schedules based on exam dates and available hours, removing the manual burden of planning.

- **Provide personalized AI-powered learning recommendations:** Offer intelligent insights on study techniques (e.g., Pomodoro, Feynman method) and dynamically adjust workloads if the student falls behind.
- **Improve productivity and learning efficiency:** Summarize lengthy lecture notes, transcripts, and study materials into digestible bullet points, significantly saving reading time.
- **Reduce academic stress and burnout:** Visually track daily workloads and proactively warn students before they overcommit their schedule.

1.5 Scope of the Project

OrbitMind AI encompasses a secure user authentication system, a responsive web dashboard, a dynamic task board, and multiple AI-driven modules (Chatbot, Study Planner, and Summarizer). The primary target audience is university students. Out of scope for this semester is integration with external university portals (like Moodle or Canvas) and native mobile application development (though the web app will be fully mobile-responsive).

CHAPTER II

System Requirements

2.1 Functional Requirements

Functional requirements define the exact behaviors, interactions, and capabilities the system must support to fulfill user needs.

User Authentication & Management

- **FR-01:** Users can register for a new account by providing their full name, email address, secure password, university name, and department.
- **FR-02:** Users can log in securely using their registered email and password to receive a session token.
- **FR-03:** Users can view, edit, and update their profile preferences at any time.

Task Management & Analytics

- **FR-04:** Users can create, view, edit, and delete tasks. Each task must support a title, description, deadline, priority level, category, and estimated completion time.

- **FR-05:** The system tracks task completion states and calculates the user's daily and weekly workload hours.
- **FR-06:** The system displays workload data visually on the student dashboard using dynamic charts.

AI Integration & Automation

- **FR-07:** Users can generate a personalized study plan by providing a subject name, syllabus topics, estimated difficulty, target exam date, and daily available study hours.
- **FR-08:** The system generates a downloadable (PDF) day-by-day study roadmap based on the AI's analysis of the user's inputs.
- **FR-09:** Users can interact with a 24/7 AI Chatbot to ask academic questions, request study strategies, or ask for motivation.
- **FR-10:** The AI Chatbot can parse natural language commands (e.g., "Remind me to study math tomorrow") and automatically create tasks on the user's task board.
- **FR-11:** Users can paste lengthy text (like lecture transcripts or articles) into the Note Summarizer and receive a short paragraph summary and key bullet points.

Notifications & Alerts

- **FR-12:** The system generates automated notifications to alert users of pending tasks that are due within the next 48 hours.
- **FR-13:** The system triggers a "burnout warning" notification if a user schedules an excessive number of high-effort tasks on a single day.

2.2 Non-Functional Requirements

Non-functional requirements describe the quality, performance, and security standards the system must meet.

Performance & Scalability

- **NFR-01: Response Time:** The backend REST API must process standard database queries and return responses in under 2 seconds.
- **NFR-02: AI Latency:** AI generation requests (summaries, study plans) must return a response within 10 seconds and display a loading indicator to the user.
- **NFR-03: Responsive UI:** The user interface must be fully responsive, adapting flawlessly to desktop, tablet, and mobile browsers.

Security & Privacy

- **NFR-04: Data Encryption:** User passwords must be securely hashed and salted using bcrypt before being stored in the database.
- **NFR-05: Session Security:** The system must use JSON Web Tokens (JWT) to manage user sessions. APIs must verify this token to guarantee data isolation (students can only access their own data).
- **NFR-06: Data Protection:** The system must sanitize all user inputs to prevent SQL/NoSQL injection and Cross-Site Scripting (XSS) attacks.

Reliability & Availability

- **NFR-07: High Availability:** The deployment platforms (Render/Vercel and MongoDB Atlas) should aim for 99% uptime during the semester.
- **NFR-08: Fault Tolerance:** The system must handle AI API rate limits gracefully, displaying user-friendly error messages or using mock-simulation fallbacks if the API is unavailable.

CHAPTER III

User Roles & Use Cases

3.1 User Roles

3.1.1 Student / User (Primary Role)

- **Responsibilities:**
  - Create, organize, and track daily academic tasks.
  - Input accurate syllabus data to generate realistic AI study plans.
  - Interact responsibly with the AI components (Summarizer, Chatbot) for academic growth.
  - Monitor personal workload statistics via the dashboard.

3.1.2 System Administrator (Secondary Role)

- **Responsibilities:**
  - Monitor overall system health, server uptime, and database performance.
  - Manage API rate limits and monitor Google Gemini/OpenAI token usage to prevent service outages.
  - Investigate and resolve user-reported bugs or system errors.

3.2 Use Cases

Use cases describe how specific actors interact with the system to achieve a goal.

Use Case 3.2.1 Secure User Registration

- **Actor:** Student
- **Action:** The student navigates to the registration page, fills out the form with their personal details, email, and a strong password, and clicks "Sign Up".
- **System Response:** The system validates the input format, verifies the email is not already in use, securely hashes the password, creates a new user record in the database, and redirects the student to the login page with a success message.

Use Case 3.2.2 AI Study Plan Generation

- **Actor:** Student
- **Action:** The student navigates to the Study Planner, inputs "Data Structures", lists 5 syllabus topics, sets the exam date for 14 days away, allocates 2 hours a day, and clicks "Generate Plan".
- **System Response:** The system packages the inputs into an engineered prompt, calls the AI API, receives a structured JSON roadmap, parses it, and displays a day-by-day calendar view of the study schedule.

Use Case 3.2.3 Interactive Academic Coaching (Chatbot)

- **Actor:** Student
- **Action:** The student types, "I am feeling really overwhelmed with my Java project, break it down for me" into the AI chat interface.
- **System Response:** The system sends the conversation history and the prompt to the AI. The AI responds with an encouraging message and a structured 4-step breakdown of the Java project. The system displays this response in the chat window.

Use Case 3.2.4 Automated Note Summarization

- **Actor:** Student
- **Action:** The student pastes a 2,000-word history lecture transcript into the Summarizer tool and clicks "Summarize".
- **System Response:** The system processes the text through the AI model, extracts the core historical events, and displays a concise 3-sentence summary alongside 5 key takeaway bullet points.

Use Case 3.2.5 Task Management and Prioritization

- **Actor:** Student
- **Action:** The student clicks "Add Task", enters "Submit Software Proposal", sets the deadline for Friday, marks it as "High Priority", and clicks Save.
- **System Response:** The system saves the task to the database, instantly updates the student's workload chart on the dashboard, and schedules an automated 48-hour reminder notification.